

Reading Time: An initial 2 minutes to view BOTH sections.



MATHEMATICS METHODS : UNITS 1 & 2, 2023

Test 2 – (10%)

1.2.5, 1.2.9 to 1.2.18, 1.2.22 to 1.2.23, 1.3.7 to 1.3.16,
1.1.4 to 1.1.11

PM

Time Allowed 20 minutes	First Name	Surname	Marks 22 marks
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Circle your Teacher's Name: Mrs Alvaro Mr Bartoshefski Mr Buckland Mr Liu
 Mr Lockyer Mr Luzuk Mr Mohammadi
 Ms Narendranathan Mr Ng
 Mr Stephens Mr Tanday

Assessment Conditions: (N.B. Sufficient working out must be shown to gain full marks)

- ❖ Calculators: Not Allowed
- ❖ Formula Sheet: Provided
- ❖ Notes: Not Allowed

PART A – CALCULATOR FREE

Question 1 [4 marks]

A cubic polynomial $f(x)$ satisfies the following conditions:

- $f(-4) = 0$,
- $f(0) = -2$ and
- it has a repeated root at $x = 1$.

Find the equation of $f(x)$ in the form $ax^3 + bx^2 + cx + d$.

Question 2 [9 marks: 2, 4, 3]

Solve the following equations:

a) $3(5 - x)(3x + 4) = 0$

b) $x = 2x^3 + 6x^2$

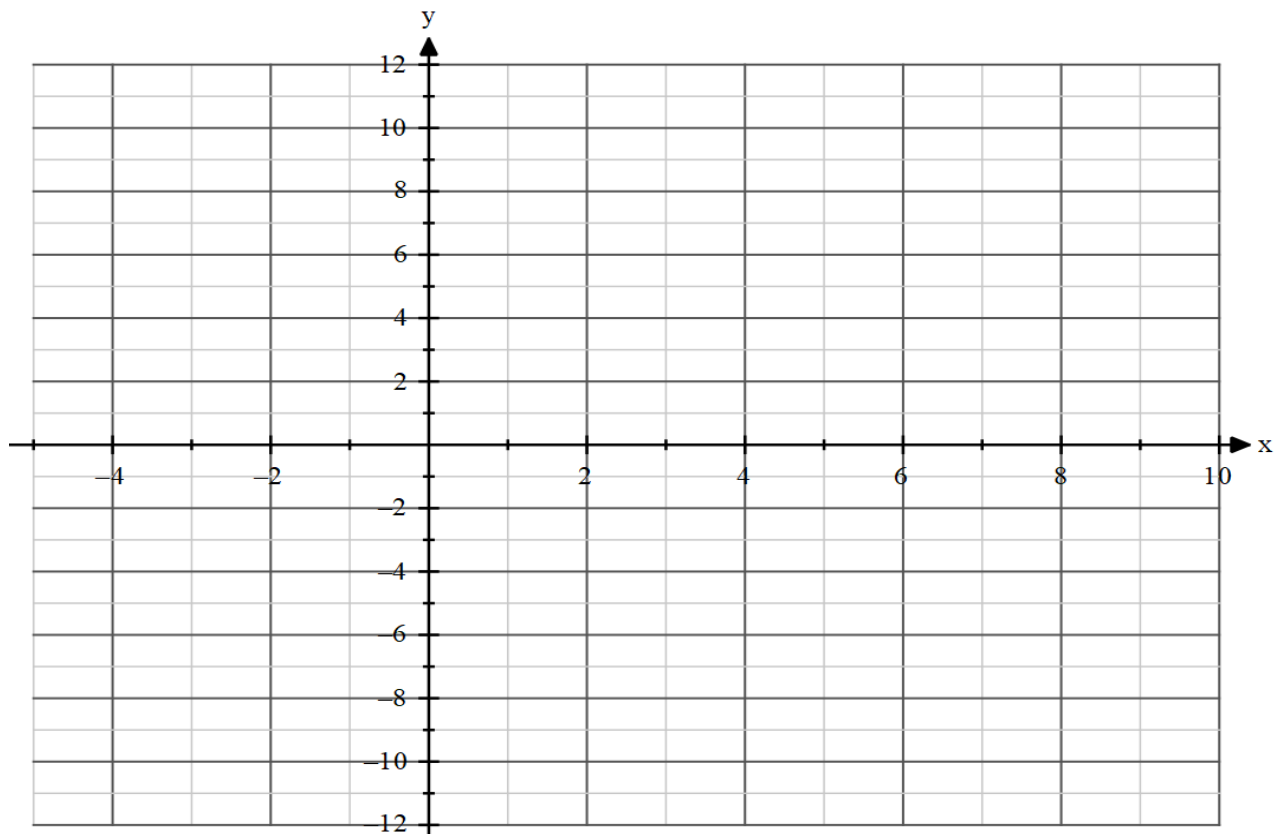
c) $\cos 2x = -\frac{\sqrt{2}}{2}$ for $0 \leq x \leq \pi$.

Question 3 [4 marks: 1, 3]

Given that $f(x) = \frac{6}{x-3}$.

a) State the equations of the asymptotes.

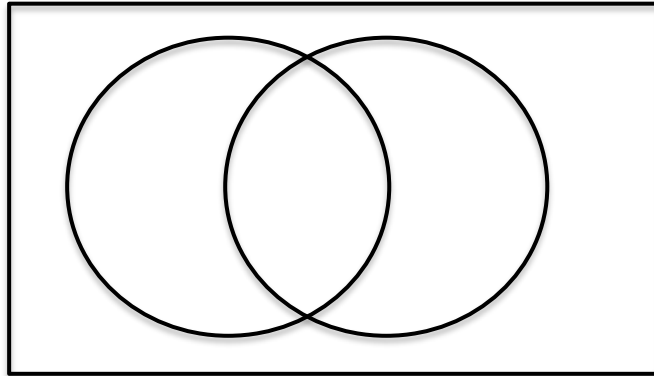
b) Sketch the graph of $f(x)$ on the axes provided, clearly labelling the key features.



Question 4 [5 marks: 2, 1, 2]

Jon has completed his Semester 1 exams for Mathematics Methods and Applications. The probability of him getting a 'B' grade in Methods is 0.2 and in Applications is 0.35. There is a 60% chance that he will not get a 'B' in either course.

- a) Represent the above information on the Venn diagram below by clearly labelling the events and placing probabilities in all regions.



- b) What is the probability that Jon will get a 'B' in both courses?
- c) What is the probability that he gets a 'B' for Methods given that he only got one 'B'?

End of Calculator Free Section



MATHEMATICS METHODS : UNITS 1 & 2, 2023

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1.2.5, 1.2.9 to 1.2.18, 1.2.22 to 1.2.23, 1.3.7 to 1.3.16,
1.1.4 to 1.1.11

PM

Time Allowed 30 minutes	First Name	Surname	Marks
			30 marks

Circle your Teacher's Name: Mrs Alvaro Mr Bartoshefski Mr Buckland Mr Liu
Mr Lockyer Mr Luzuk Mr Mohammadi
Ms Narendranathan Mr Ng
Mr Stephens Mr Tanday

Assessment Conditions: (N.B. Sufficient working out must be shown to gain full marks)

- ❖ Calculators: Allowed
- ❖ Formula Sheet: Provided
- ❖ Notes: Not Allowed

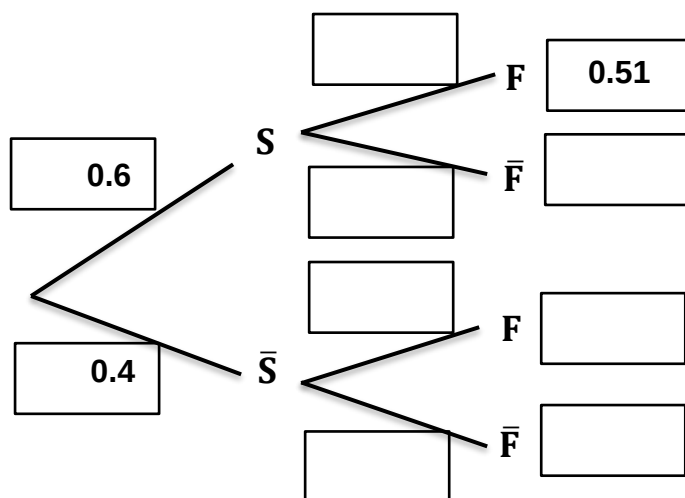
PART B – CALCULATOR ASSUMED

All approximate answers to be stated to an accuracy of 2 decimal places, unless otherwise directed.

Question 5 [7 marks: 3,4]

A survey about Year 12 leavers travel plan showed that 60% had an overseas destination (event S) and that 65% had travelled with friends (event F). Also, 51% of the leavers had travelled overseas with friends.

a) Use the above information to complete all the probabilities in the tree diagram.



b) Determine the probability that a randomly selected Year 12 leaver:

(i) Travelled with friends given that they did not travel overseas.

(ii) Travelled with friends or overseas.

(iii) Did not travel overseas given that they travelled with friends, leaving your answer as a simplified fraction or correct to 4 decimal places.

Question 6 [4 marks]

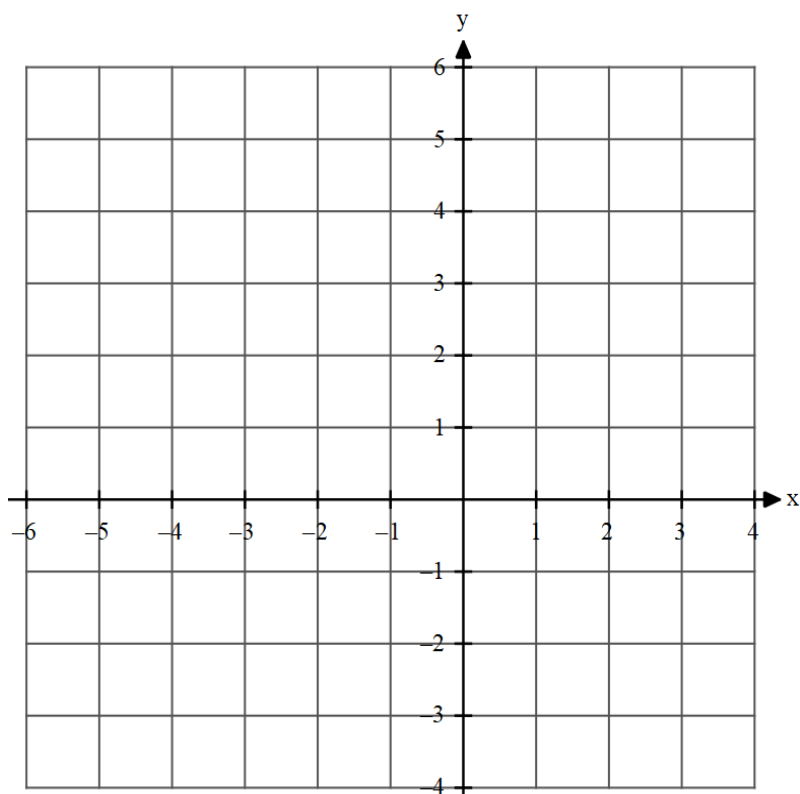
If $\sin X = -\frac{\sqrt{7}}{4}$ and $\cos Y = \frac{21}{29}$ with X and Y in the same quadrant, find, leaving your answer in exact value, the value of $\cos(X + Y)$.

Question 7 [4 marks: 2, 2]

The equation of a circle is given by $x^2 + y^2 + 2x - 4y = 4$.

a) Find the centre and radius of this circle.

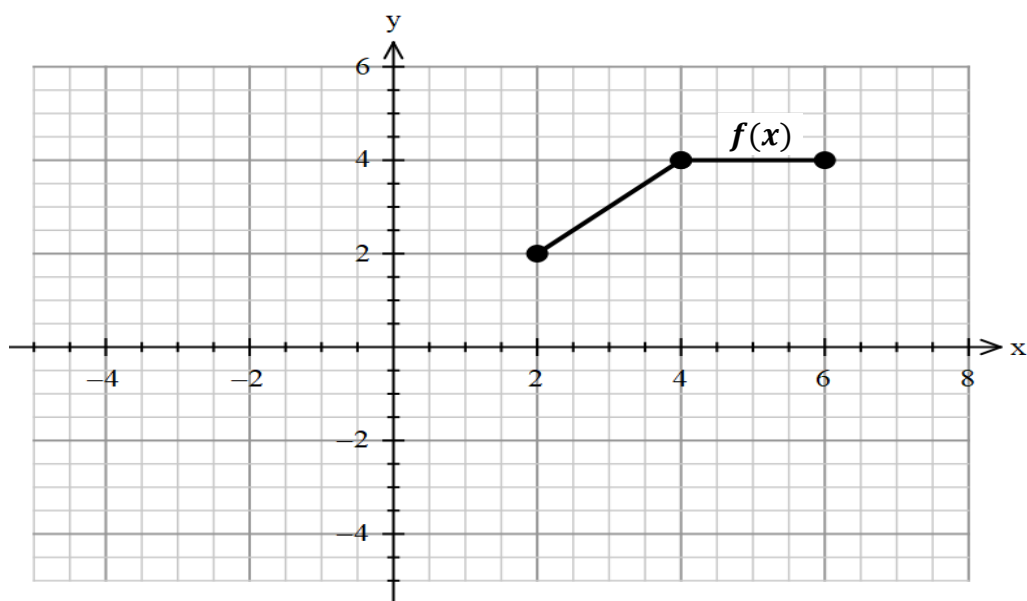
b) Sketch the circle on the grid given, indicating the centre and radius.



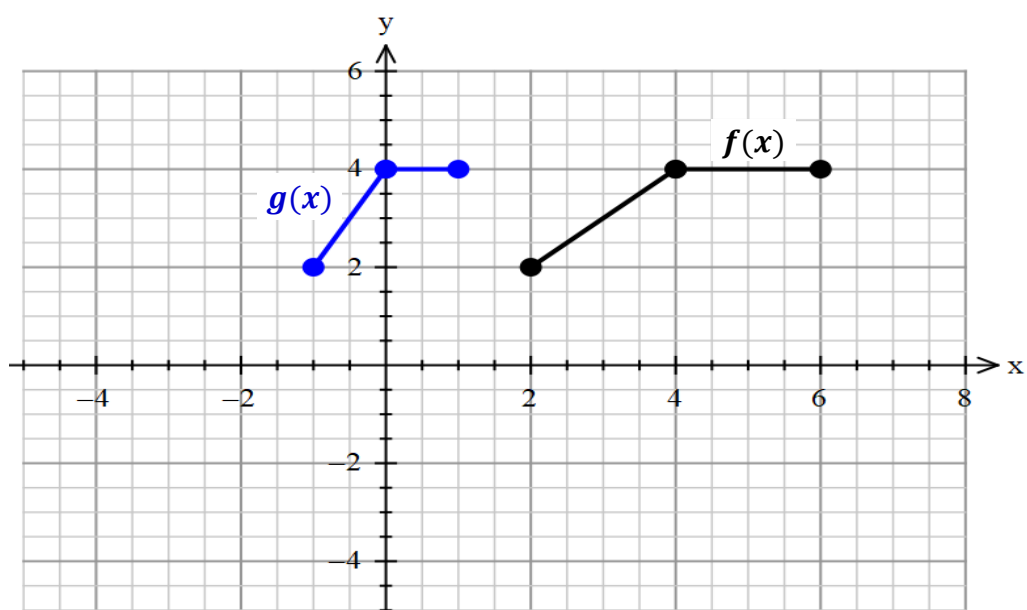
Question 8 [6 marks: 2,2,2]

The graph of the function $y = f(x)$ is shown below.

- a) Draw, on the same grid, the graph of $y = \frac{1}{2}f(x) - 3$.



- b) $f(x)$ and $g(x)$ are graphed below.



- (i) State the transformations, in order, that map $f(x)$ to $g(x)$.

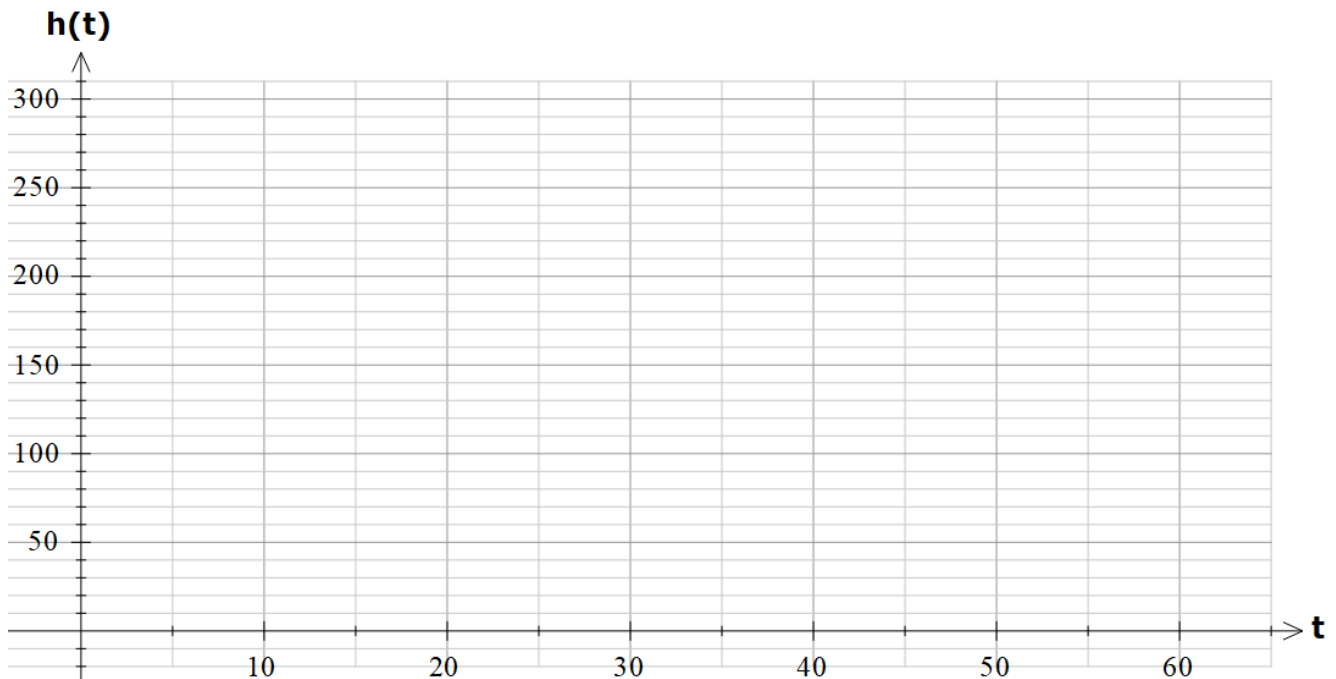
- (ii) State $g(x)$ as a function of $f(x)$.

Question 9 [9 marks: 1,3,1,2,2]

The Dubai Eye is the largest Ferris wheel in the world, and it moves in a vertical circle. To provide a different experience for the passengers, the function of motion has been re-configured to

$h(t) = 125 \left(1 - \cos \left(\frac{\pi}{15} t \right) \right)$, where h measures the height above the ground in metres and t is the time in minutes since the measurements began.

- a) Determine the maximum height that can be reached by the Dubai Eye.
- b) Sketch the graph of $h(t)$ on the axes below for $0 \leq t \leq 60$.



- c) What is the period of the motion?
- d) Starting from the ground level, Jerry had an hour's ride on the Dubai eye. Determine the time(s) where he is at the highest point.
- e) Tom completed one revolution of the ride where he also started at the ground level. What was the total duration that he was at a height of at least 180 metres above the ground during his ride?

End of Calculator Assumed Section